

Qiwen Xiao

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Education

Southern University of Science and Technology of China (SUSTech) B.Eng. in Computer Science and Technology	Sep 2022 – (Jun 2026)
University of California, Irvine 3+2 Engineering Program, Henry Samueli School of Engineering	Sep 2025 – (2027)
University of California, Davis Global Study Program (Summer Sessions I & II)	Jun 2024 – Sep 2024

Research

Undergraduate Researcher , HPC Forge @ UC Irvine - Advised by Prof. Aparna Chandramowlishwaran under EECS199, Winter 2026. - Studied fused W4A16 INT4 weight-only GEMM in Triton for LLM inference. - Implemented and benchmarked fused data-parallel and split-K kernels for decode and prefill regimes.	Jan 2026 – Mar 2026
Undergraduate Research Assistant , Computer Vision Lab @ UC Irvine - Advised by Prof. Glenn Healey under EECS199, Fall 2025. - Aiming at wild fire prevention, utilized XGBoost to discover the relationship between radiance spectrum and water thickness of various vegetation. - Leveraged Levenberg–Marquardt algorithm and PROSPECT leaf model to find the best leaf parameters for a reflectance spectrum.	Sep 2025 – Jan 2026
Undergraduate Researcher , AI Research Institute @ Shenzhen MSU-BIT University - Advised by Prof. Xitong Li and Prof. Shijie Zhang. - Developed as first author StyleMeta – a novel meta network that combines flow matching with dynamic parameter generation to rapidly produce high-quality, multi-style graphical layouts while significantly reducing training time and computational overhead (paper under review at ICWS'26).	Sep 2024 – Aug 2025
Undergraduate Researcher , ARiSE Lab @ SUSTech - Advised by Prof. Yuqun Zhang. - Leveraged entropy information to evaluate patch correctness and facilitate ranking of patches generated by large language models trained on code. - Engaged in training a large language model on code using Megatron, theoretically and empirically discovering that dependency context facilitates convergence.	Jan 2024 – Jul 2025
Member , Supercomputing Team @ SUSTech - Advised by Prof. Zhuozhao Li. - Participated in supercomputing competitions including ASC24, APAC24, IndySCC24, ISC25, and ICT25. - Exhibitor at SC24. - Co-organized the 2nd SUSTech SuperComputing Competition and served as the task maker of optimizing CloverLeaf, a mini-app for high-performance computing (Website).	Sep 2023 – Sep 2025

Projects

Physics-Guided Conditional Flow Matching for Lensless Imaging , 2026 - The first to formulate lensless reconstruction as conditional generation and study its empirical advantages over optimization-based and supervised baselines. - Compared two practical parameterizations, velocity prediction and image prediction with induced velocity. - Final project of <i>EECS298: Computational Optics</i> at UC Irvine.	Charley-xiao/lensless-flow
ODE Samplers for Rectified Flow Generative Models , 2025 - Experimentally and theoretically compared six different ODE samplers in the context of rectified flow generative models and discussed various trade-offs and which one to use given certain constraints. - Final project of <i>CS274E: Deep Generative Models</i> at UC Irvine.	Charley-xiao/cs274e-project
Boston in 3D: A Citywide Object Archive Using Nerfstudio , 2025 - Led a team to reconstruct objects across the city of Boston using Nerfstudio and to develop an interactive map. - Final project of <i>MIT Blended Learning</i> at MIT.	Charley-xiao/NVIDIA_MIT
Explainable Writing Style Detection , 2024 - Developed a novel pipeline for writing style detection and authorship attribution, which combines a fastText-based text classifier with SHAP and LIME for model explainability and features real-time visualization. - Final project of <i>ECS171: Machine Learning</i> at UC Davis.	Charley-xiao/ecs171-project

- Built a cross-platform software application as your personal manager for campus events and reservations.
- Final project of *CS304: Software Engineering* at SUSTech.

Skills

Programming Languages: C++, C, Java, Python, SQL, JavaScript, MATLAB, Kotlin,...

Technologies: PyTorch, Transformers, Diffusers, Vue3, Tornado, React.js,...

Human Languages: Mandarin (Native), English (C1), Spanish (B1).

Awards

Second Prize , Asian Supercomputing Challenge (ASC)	2024
Second Prize , The 2nd SUSTech Quantitative Trading Competition	2023
Grand Prize , NECCS (National English Competition for College Students)	2023, 2024
Second Prize , National Olympiad for Informatics (Provincial)	2019, 2020